

LISA CUNEO

PERSONAL INFORMATION

Place of Birth	Genova, Italy
Date of Birth	25 July 1995
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PERSONAL INTRODUCTION

My studies have focused on applied mathematics. My Bachelor and Master thesis concerned inverse problems, in particular the spatial resolution of Electro/Magneto-encephalography and, a widely used brain image techniques. I took part to several Schools on biomedical data analysis and Machine Learning. During the two stages at Axpo, I have increased my skills of data analysis, developing algorithms for anomaly detection via model design. I spent six months at the Department of Neuroscience and Biomedical Engineering (NBE) at Aalto University, Helsinki (Finland), within the Erasmus+ program, to work on my Master thesis. I taken the Phd in Physics and nanosciences - bio-nanosciences curriculum. During the period abroad in Cambridge (UK) I had the opportunity to solidify my understanding of common deep learning architectures and acquire new skills in the field of data analysis and generative models. Now I'm undertaking a Post Doc at the IIT between the Molecular Microscopy and Spectroscopy and the Computational Statistics and Machine Learning groups.

TECHNICAL EXPERTISE

- Biomedical data
- Data analysis
- Computer Science
- Inverse Problem
- Machine Learning
- Microscopy imaging

PROFESSIONAL EXPERIENCE

<i>Stage Axpo, Risk Management</i>	<i>Jun - Aug 2019</i> AXPO – Genova, Italy Analysis of transactions to automatically check if their modification processes have been carried out in compliance with the policies. The stage was focused on pattern recognition, model design, anomaly detection. Reference: Ing. Fabrizio PISCHEDDA · fabrizio.pischedda@axpo.com
<i>C.Co.C. INSPIRE S.R.L, Start Up</i>	<i>Jan - Jun 2019</i> INSPIRE S.R.L – Genova, Italy Applied scientific research that aims to create and test an innovative land monitoring system, through the use of drones. Creation of Matlab software for testing and demonstrating the identified algorithm. Search for the most efficient constrained scheduling solution for multiple requests. Reference: Ing. Marco GHIO · m.ghio@be-inspire.com
<i>Internship Axpo, Risk Management</i>	<i>Mar - May 2019</i> AXPO – Genova, Italy Validation of commodity prices coming from a model. Implementation of an algorithm to detect anomalies in historical series of forward market prices. Reference: Ing. Fabrizio PISCHEDDA · fabrizio.pischedda@axpo.com

EDUCATION

<i>Post Doc in Microscopy and Machine Learning</i>	2024-On going	Istituto Italiano di Tecnologia (IIT) – Genova, Italy I'm part of the <i>Molecular Microscopy and Spectroscopy</i> and <i>Computational Statistics and Machine Learning</i> groups of the Istituto Italiano di Tecnologia (IIT) under the supervision of Giuseppe VICIDOMINI and Massimiliano PONTIL.
<i>PhD in Physics and Nanosciences</i>	2020-2024	Università degli studi di Genova & IIT – Genova, Italy I was part of the <i>Nanoscopia</i> & NIC@IIT group of the Istituto Italiano di Tecnologia under the supervision of Prof. Alberto DIASPRO. Graduation date: 16/05/2024 My reasearch is focused on computational analysis of biomedical images.
<i>Visitor student</i>	May - Sep. 2023	University of Cambridge – Cambridge, United Kingdom Visitor student at the Computational Biology at Department of Computer Science and Technology under the supervision of Prof. Pietro Liò.
<i>Erasmus+ Traineeship</i>	Sep. 2019 - Feb. 2020	AALTO UNIVERSITY – Helsinki, Finland Visitor student at the Neuroscience and Biomedical Engineering (NBE) department under the supervision of Prof. Lauri PARKKONEN.
<i>PERCORSO FORMATIVO 24 CFU</i>	2019-2020	Università degli studi di Genova – Genova, Italy Course to obtain the 24 credits in the anthro-po-psycho-pedagogical disciplines and in the methodologies and educational technologies required for participation in the competition for public schools as per Ministerial Decree 257/2017.
<i>Masters of Applied Mathematics</i>	2017-2020	Università degli studi di Genova – Genova, Italy Thesis Title: <i>An inverse problem perspective on the optimal design of sensor arrays for on-scalp MEG.</i> Supervisors: Prof. Alberto SORRENTINO and Prof. Lauri PARKKONEN Final degree mark: $^{110}/_{110}$ cum laude Graduation date: 25/03/2020
<i>Bachelor of Mathematics</i>	2014-2017	Università degli studi di Genova – Genova, Italy Thesis Title: <i>Numeric study of spatial resolution of EEG through ordinary least squares problems.</i> Supervisor: Prof. Alberto SORRENTINO Final degree mark: $^{102}/_{110}$ Graduation date: 15/11/2017

TEACHING ASSISTANT

<i>Mathematical Analysis module 2</i>	2021-2023	Università degli studi di Genova – Genova, Italy Exercise classes for "Mathematical Analysis module 2" course, cod. 60235 SSD MAT/05, I semester, year 2 of the Degree in Mechanical Engineering Supervisor: Prof. Edoardo MAININI
<i>Statistics</i>	2021-2023	Università degli studi di Genova – Genova, Italy Exercise classes for "descriptive statistics" course, cod. 52480 SECS - S/01, II semester, year 1 of the Degree in mathematics and SMID Supervisor: Prof. Alberto SORRENTINO and Francesco PORRO
<i>Mathematical Analysis 2</i>	2022-2023	Università degli studi di Genova – Genova, Italy Exercise classes for "Mathematical Analysis " course, cod. 25900 SSD MAT/05, I semester, year 2 of the Degree in Mathematics Supervisor: Prof. Giovanni ALBERTI

REVIEW ACTIVITY

Reviewer for:

- Microscopy Research and Technique (MRT), Wiley

PUBLICATIONS

- Published* Medical Image Analysis
Hunting imaging biomarkers in pulmonary fibrosis: Benchmarks of the AIIB23 challenge.
Guang YANG, Michail MAMALAKIS, Lisa CUNEO, Francesco PRINZI, Gianluca CARLINI et all
DOI: <https://doi.org/10.1016/j.media.2024.103253>.
- Published* Biophysical Journal
Remodeling of the nuclear landscapes and epigenome during retro-transformation of neuroblastoma cells mediated by the non-coding RNA NDM29
Francesca BALDINI, Lama ZEAITER, Lisa CUNEO, Paolo BIANCHINI, Laura VERGANI, Aldo PAGANO, Alberto DIASPRO
DOI: <https://doi.org/10.1016/j.bpj.2023.11.561>.
- Published* Biochimica et Biophysica Acta (BBA) - Molecular and Cell Biology of Lipids
Nuclear and chromatin rearrangement associate to epigenome and gene expression changes in a model of in vitro adipogenesis and hypertrophy.
Francesca BALDINI, Lama ZEAITER, Farah DIAB, Hawraa ZBEEB, Lisa CUNEO, Aldo PAGANO, Piero PORTINCASA, Alberto DIASPRO, Laura VERGANI
DOI: <https://doi.org/10.1016/j.bbalip.2023.159368>.
- Published* IL NUOVO CIMENTO
An automated tool to estimate chromatin compaction in stained nuclei.
Lisa CUNEO, Marco CASTELLO, Francesca BALDINI, Alberto DIASPRO
DOI: <https://doi.org/10.1393/ncc/i2022-22193-5>.
- Published* IL NUOVO CIMENTO
Investigating nanoscale chromatin alterations involved in neuroblastoma transformation by optical nanoscopy.
Francesca BALDINI, Isotta CAINERO, Lisa CUNEO, Michele ONETO, Elena GATTA, Chantal USAI, Aldo PAGANO, Laura VERGANI, Alberto DIASPRO
DOI: <https://doi.org/10.1393/ncc/i2022-22191-7>.
- Published* IL NUOVO CIMENTO
A deep learning-based method to spectrally separate overlapping fluorophores based on their fluorescence lifetime.
Lisa CUNEO, Marco CASTELLO, Simonluca PIAZZA, Irene NEPITA, Isotta CAINERO, Giorgio TORTAROLO, Luca LANZANÒ, Paolo BIANCHINI, Giuseppe VICIDOMINI, Alberto DIASPRO
DOI: <https://doi.org/10.1393/ncc/i2023-23130-x>.
- Published* La rivista del nuovo cimento
Emerging Mueller matrix microscopy applications in biophysics and biomedicine.
Alberto DIASPRO, Paolo BIANCHINI, Fabio CALLEGARI, Lisa CUNEO, Riccardo MARONGIU, Aymeric LE GRATIET, Ali MOHEBI, Marco SCOTTO, Colin J. R. SHEPPARD
DOI: <https://doi.org/10.1007/s40766-023-00046-5>.
- Conference proceedings* EPJ Web Conf. Volume 287, EOS Annual Meeting (EOSAM), September 2023
The Artificial Microscope.
Alberto DIASPRO, Paolo BIANCHINI, Lisa CUNEO
DOI: <https://doi.org/10.1051/epjconf/202328713012>.

Conference proceedings	The Italian Physical Society - Proceedings of the International School of Physics "Enrico Fermi" - Volume 210: Multimodal and Nanoscale Optical Microscopy - pp. 143 – 145 - September 2023 <i>Scattering Networks: a tool to remove the background.</i> Lisa CUNEO, Simone CIVITA, Paolo BIANCHINI, Alberto DIASPRO DOI: https://doi.org/10.3254/ENFI230014.
Conference proceedings	Biophysical Journal 122 (3) pp. 462a February 2023 <i>A deep learning method to separate fluorophores based on their fluorescence lifetime.</i> Lisa CUNEO, Marco CASTELLO, Simonluca PIAZZA, Irene NEPITA, Luca LANZANÒ, Paolo BIANCHINI, Giuseppe VICIDOMINI, Alberto DIASPRO DOI: https://doi.org/10.1016/j.bpj.2022.11.2483.
Conference proceedings	Biophysical Journal 121 (3) pp. 476a February 2022 <i>Decrypting nanoscale chromatin architecture alterations implicated in neuroblastoma transformation by optical nanoscopy.</i> Francesca BALDINI, Isotta CAINERO, Lisa CUNEO, Irene NEPITA, Chantal USAI, Paolo BIANCHINI, Laura VERGANI, Aldo PAGANO, Alberto DIASPRO DOI: https://doi.org/10.1016/j.bpj.2021.11.402.
Conference proceedings	Biophysical Journal 121 (3) pp. 362a February 2022 <i>Investigating the role of chromatin compaction at the nanoscale in Hutchinson-Gilford progeria syndrome using expansion microscopy</i> Chantal USAI, Isotta CAINERO, Lisa CUNEO, Francesca BALDINI, Matteo MARIANGELI, Irene NEPITA, Paolo BIANCHINI, Alberto DIASPRO DOI: https://doi.org/10.1016/j.bpj.2021.11.938.
Conference proceedings	Technical Digest Series (Optica Publishing Group), paper JTh5A.111 November 2021 <i>An Automated Tool to Analyse 3D Fluorescence Images of Stained Nuclei.</i> Lisa CUNEO, Francesca BALDINI, Marco CASTELLO, Irene NEPITA, Simonluca PIAZZA, Laura VERGANI, Alberto DIASPRO DOI: https://doi.org/10.1364/FIO.2021.JTh5A.111.
Accepted	Scientific Reports <i>Advanced Background Removal Methods in Single Molecule Localization Microscopy Using Scattering Networks and SVD</i> Lisa CUNEO, Simone CIVITA, Ivan TRAPASSO, Luca RATTI, Paolo BIANCHINI, Alberto DIASPRO
Under review	Medical Image Analysis <i>Self-Tuning Regularization for Image Scanning Microscopy</i> Sofia AGOSTONI, Lisa CUNEO, Christian DANIELE, Giacomo GARRÈ, Alessandro ZUNINO, Luca CALATRONI, Giuseppe VICIDOMINI Arxiv: https://doi.org/10.48550/arXiv.2605.31426.
In preparation	Nature Methods <i>Independent noise realizations enable morphologically agnostic image reconstruction in single-photon-sensitive microscopy</i> Lisa CUNEO, Alessandro ZUNINO, Laurent LE, Sofia AGOSTONI, Saverio SALZO, Massimiliano PONTIL, Luca CALATRONI, Giuseppe VICIDOMINI Bioarxiv: https://doi.org/10.64898/2026.09.12.751176.
In preparation	SIAM Imaging <i>Regularization by Independent Noise Realizations: A Variational Model and a Convergent Relaxed Block Proximal-Gradient Method</i> Lisa CUNEO, Saverio SALZO, Giuseppe VICIDOMINI, , Luca CALATRONI, Massimiliano PONTIL

AWARDS

Challenge	MICCAI AIIB23-Task 1 Segmentation: Airway-Informed Quantitative CT Imaging Biomarker for Fibrotic Lung Disease. Position: 8/20. Link: Task1
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Challenge MICCAI AIB23-Task 2 Binary classification: Airway-Informed Quantitative CT Imaging Biomarker for Fibrotic Lung Disease. Position: 4/6.
Link: [Task2](#)

Travel Grant SIBPA (Società Italiana di Biofisica Pura e Applicata) grant for 1500€ to participate at the BPS (BioPhysical Society) annual meeting at San Diego, California, USA.

Best communication The communication with title "An automated tool to estimate chromatin compaction in stained nuclei" was selected by the Scientific Committee for the Best Communications of 107° Congresso SIF.

Best communication The communication with title "Scattering Networks: a tool to remove the background" was selected by the Scientific Committee for the Best Flash talk of International School of Physics "Enrico Fermi" summer school "Multimodal and Nanoscale Optical Microscopy".

Best communication The communication with title "A deep learning-based method to spectrally separate overlapping fluorophores based on their fluorescence lifetime." was selected by the Scientific Committee for the Best Communications of 108° Congresso SIF.

INVITED PRESENTATIONS

Talk *March 7th, 2024* *PoliTO – Torino, Italy*
Big Data and Data Science Laboratory, DISMA, SmartData@PoliTO
Organized by: Ivan S. Trapasso.
Title: Scattering Networks and Singular Values Decomposition: different methods to remove background in Single-Molecule Localization Microscopy images.

Talk *July 21th, 2023* *CL – University of Cambridge, United Kingdom*
Artificial Intelligence Research Group Talks at the Computer Laboratory
Organized by: Prof. Pietro Liò.
Title: Scattering Networks and Singular Values Decomposition: different methods to remove background in Single-Molecule Localization Microscopy image.

Talk *June 5-6, 2023* *Stem Cell Institute – Cambridge, United Kingdom*
Cambridge Stem Cell Institute Annual Retreat, Industry Flash Talk.
Presentation of Genoa Instruments PRISM product.

Talk *May 23th, 2023* *CEB – University of Cambridge, United Kingdom*
Laser Analytics Group at the Department of Chemical Engineering and Biotechnology (CEB), Physics Troublemaking Meeting
Organized by: Prof. Clemens Kaminski.
Title: Scattering Networks and Singular Values Decomposition: different methods to remove background in Single-Molecule Localization Microscopy image.

Seminar *Oct 12th, 2020* *Inverse Problems Seminar – Online*
Inverse Problems Seminars.
Organized by: Inverse Problem group of Department of Mathematics and Statistics - University of Helsinki.
Presentation of my Master Thesis, whose topic was the optimization of an hypothetical on-scalp Magnetoencephalograph sensors configuration in terms of sensitivity and source localization.

COMPUTER SKILLS

Advanced: MATLAB, Python, R, L^AT_EX
Intermediate: Office, SQL
Basic: C++, Java

LANGUAGE SKILLS

Mother tongue: ITALIAN
B2: ENGLISH

SOFT SKILLS

- *Abstract Thinking*
- *Problem Solving*
- *Lateral Thinking*
- *Teamwork*

SCHOOLS AND WORKSHOPS

CIL School 2026 *23 - 27 Feb, 2026 – Genova, Italy*
Advanced Microscopy practical workshop.
Personal contribution: I collaborated in the organization of the event and give a talk with title "Image Scanning Microscopy: from image formation to inverse problem".
Website: [CIL](#).

Nikon Workshop 2024 *2 - 6 Dec, 2024 – Genova, Italy*
Computational Imaging & Learning: From physics to data-driven methods.
Personal contribution: I collaborated in the organization of the event.
Website: [CIL2026](#).

MALGA school *Sep. 2-6, 2024 – Genova, Italy*
Applied Harmonic Analysis and Machine Learning school, organized by: Machine Learning Genoa Center (MALGA)- Università degli Studi di Genova.
Website: [ahaml2024](#).

Nikon Whorkshop 2023 *Nov. 27 - 1 Dec, 2023 – Genova, Italy*
8th NIC@IIT Advanced Microscopy practical workshop.
Personal contribution: I collaborated in the organization of the event.
Website: [NIC@IIT2023](#).

Nikon Whorkshop 2022 *Nov. 28 - 2 Dec, 2022 – Genova, Italy*
7th NIC@IIT Advanced Microscopy practical workshop.
Personal contribution: I collaborated in the organization of the event.
Website: [NIC@IIT2022](#).

MALGA school *Sep. 5-9, 2022 – Genova, Italy*
Applied Harmonic Analysis and Machine Learning school, organized by: Machine Learning Genoa Center (MALGA)- Università degli Studi di Genova.
Website: [ahaml2022](#).

PRIMO Workshop 2022 *Sep. 6 - 8, 2022 – Trieste, Italy*
Post-graduate Researchers in Inverse problems, Machine learning, and Optimization.
I presented a talk with title "Scattering Networks: a tool to remove background in microscopy images".
Website: [PRIMO](#).

School of Physics "Enrico Fermi" *Jul. 10-15, 2022 – Varenna, Italy*
Multimodal and Nanoscale Optical Microscopy.
I presented a flash talk with title "Scattering Networks: a tool to remove the background".
Website: [Varenna](#).

UMI-MIVA Winter School 2022 *Jan. - Feb., 2022 – Online*
Advanced methods for mathematical image analysis.
Website: [MIVA](#).

Nikon Workshop *Nov. 30 - Dec. 3, 2021 – Genova, Italy*
6th NIC@IIT pratical workshop on advanced microscopy.
Website: [NIC@IIT2021](#).

VaMOS Workshop *Nov. 22 - 23, 2021 – Online*
Advanced optimization methods for inverse problems & applications to image microscopy.
Website: [VaMOS](#).

PRIMO Workshop 2021 *Oct. 11 - 13, 2021 – Online*
Post-graduate Researchers in Inverse problems, Machine learning, and Optimization.
Website: [PRIMO](#).

NonInvasive Mathematics Workshop

Apr. 13 - 16, 2021 – Online

Workshop bridging between mathematicians, engineers, neuroscientists, and clinicians working with magneto/electro-encephalography and similar techniques.

Website: [DIMA](#).

"Convex Optimization" School

Jul. 20 - 24, 2020 – Online

Organized by: Machine Learning Genoa Center - Università degli Studi di Genova.

Description: 20 hours advanced convex optimization course including theory classes and practical laboratory sessions.

"Deep Learning: A Hands-on Introduction" School *Jul. 13 - 17, 2020 – Online*

Organized by: Machine Learning Genoa Center - Università degli Studi di Genova.

Description: 20 hours advanced deep learning course including theory classes and practical laboratory sessions.

School of Neuroengineering

Jun. 18-22, 2018 – Genova, Italy

7th Summer School of Neuroengineering 'Massimo Grattarola'.

Website: [Neuroengineering](#).

Workshop "Invasive mathematics"

May. 11th, 2018 – Genova, Italy

Workshop bridging between mathematicians, engineers, neuroscientists, and clinicians working with invasive brain imaging such as StereoEEG.

OTHER ACTIVITY

Projects

During my studies I carry out the following projects:

- *Convex Optimization on Riemannian Manifolds*
Final project for *Convex Optimization* course, 2021
- *Optogenetics*
Final project for *Biophysics* course, 2021
- *Clustering of ElectroCorticoGraphy (ECoG) data during a visual categorization task*
Final project for *Machine Learning* course, 2019
- *Curvatures and their properties*
Final project for *Discrete Surface Analysis Methods and their applications* course, 2019
- *Numerical resolution of an elliptical PDE on bounded domain*
Final project for *Numerical Treatment of Differential Equations* course, 2018

September 30, 2026

SIGNATURE

Lisa Cuneo